THE NATIONAL UNIVERSITY OF LESOTHO

Department of Mathematics and Computer Science

CS3520 - Computer Organisation and Architecture I

Class Test 1

Date: 23 October 2024 Marks: 50

1. A processor executes a program containing 500 million instructions with a CPI of 2.2. The clock frequency of the processor is 2.5 GHz.

   (a) Use the CPU performance equation to calculate the total execution time for the program. [4] (b) If you optimize the program to reduce the CPI to 1.8, how much faster (as a percentage) will the optimized program run compared to the original? [4]

   (c) Discuss one scenario where CPI optimization would be preferred over increasing clock frequency. [4]

2. You are working with a 16-bit processor supporting several instruction types, including arithmetic, load/store, and branching instructions. The processor's instruction set is limited to 16-bit instructions, and each instruc tion includes two operands with 6-bit operand speci ers.

   (a) What is the maximum number of two-operand instructions that the instruction set can support? [3] (b) If the system must support at least 30 one-operand instructions and 15 zero-operand instructions, explain how you would design the instruction formats to t within the 16-bit limit. [4]

   (c) Discuss how adding more operands in instructions impacts the complexity of both hardware and software design. [3]

3. A C program performs the following operation:

   A = B + (C < < 2);

   (a) Write the equivalent MIPS assembly instructions to implement the above operation, assuming the vari ables are stored in registers: $s0 = A, $s1 = B, $s2 = C. [5]

   (b) Explain why shift operations (like < <) are often preferred over multiplication in low-level programming. [3]

4. Expanding the MIPS instruction set to 128 registers and adding four times as many instructions requires adjustments to the instruction encoding.

   (a) How would the size of R-type and I-type bit elds change to accommodate the expanded instruction set? [5]

   (b) Discuss one advantage and one disadvantage of expanding the instruction set in this way. [5]

5. The IBM 370 architecture does not support indirect addressing. Assume you must retrieve an operand stored at address 800, but the address is rst accessed indirectly via memory location 400.

   (a) Write a sequence of instructions (in pseudocode or MIPS) to access the operand stored at address 800 using the IBM 370 approach. [5]

   (b) Explain how the absence of indirect addressing in uences performance and code complexity in real-world applications. [5]